

MACULAR HOLE SURGERY AND RETINAL TECTONICS

The Impact of Internal Limiting Membrane Peeling Size on Tangential Retinal Displacement

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Purpose: To evaluate the tangential retinal displacement occurring following macular hole (MH) surgery and to assess the impact of the internal limiting membrane (ILM) peeling size on the extent of the retinal movement.

Methods: This retrospective study included patients with full-thickness MH undergoing 25-gauge pars plana vitrectomy with ILM peeling. Patients received either a small ILM peeling with a size of two-disk diameters or a large peeling extended up to the vascular arcades. Near-infrared retinal imaging was performed with the Spectralis (Heidelberg Engineering, Carlsbad, Germany) before and 6 months after the surgery. The tangential retinal displacement was evaluated comparing the optical flow of near-infrared images with a custom digital image analysis algorithm.

Results: Forty-four eyes of 44 patients undergoing vitrectomy with small ($n = 24$) or large ($n = 20$) ILM peeling were included. An average overall displacement of $31.3 \pm 22.8 \mu\text{m}$ toward the optic disk was observed after the surgery. Large ILM peeling was associated with a significantly higher overall displacement ($P = 0.009$), displacement in the central 4-mm circle ($P < 0.001$), and displacement in outer 8-mm ring ($P = 0.001$). Macular holes closure was achieved in 100% and 83.3% of patients in the large and small peeling group, respectively ($P = 0.055$).

Conclusion: Pars plana vitrectomy with ILM peeling for MH results in a tangential retinal displacement toward the optic disk. A larger extent of the ILM peeling leads to a greater tangential movement, possibly improving the MH closure rate.

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Initial reports on the surgical management of full-thickness macular holes (MHs) published in 1991 documented success rates of 58% following pars plana vitrectomy alone.¹ Since then, the surgical technique has evolved to include peeling of the internal limiting membrane (ILM), gas tamponade, and post-

operative face-down positioning, achieving closure rates as high as 90% to 100%.^{2–4} Therefore, ILM peeling has now become an integral part of the gold-standard treatment for full-thickness MHs.

Although the ILM has a thickness of only 0.4 to 1.4 μm , its mechanical strength is approximately 1,000 times higher than other retinal cellular layers, contributing to more than 50% of the overall retinal rigidity.^{5,6} Therefore, removal of the ILM releases the retinal inner tensions, resulting in a tangential displacement of the tissue, which allows for the closure of the hole.⁷ To date, however, it remains unclear whether the extent of ILM peeling may influence the magnitude and direction of retinal movement and MH closure rate.

Scarinci et al⁸ recently described a novel method to assess the “en face” retinal displacement occurring

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after pars plana vitrectomy by comparing the optical flow of sequential near-infrared retinal photographs. A significant tangential displacement correlating with the visual function recovery was observed in eyes undergoing epiretinal membrane surgery. In this study, we employed a modification of the image analysis method to measure the tangential retinal displacement occurring following MH surgery. The impact of the ILM peeling extent on the tangential movement was also investigated.

Methods

This retrospective study included patients who underwent pars plana vitrectomy for full-thickness MH at the Ophthalmology Unit of the Sant'Anna University Hospital (University of Ferrara, Ferrara, Italy) between January 2022 and December 2023. Exclusion criteria were as follows: other retinal diseases such as diabetic retinopathy or age-related macular degeneration; secondary MH; MH retinal detachment; high myopia (ocular axial length >26.00 mm); and lack or poor quality of near-infrared retinal imaging. Macular holes were classified according to the scheme proposed by Gass.⁹ The study adhered to the tenets of the 2013 Declaration of Helsinki and was approved by the local institutional review board.

Surgical Technique

All patients were operated by two surgeons with the same surgical technique except for the ILM peeling extent. In one case, a standard peeling with a width of approximately two-disk diameters was performed (small peeling group); in the other, peeling was extended up to the vascular arcades and temporal margin of the optic nerve head (large peeling group). The remaining surgical steps are summarized as follows: standard phacoemulsification and intraocular lens (IOL) implantation was performed in all phakic patients. Twenty-five-gauge pars plana vitrectomy was performed with the Constellation Vision System (Alcon Laboratories Inc, Fort Worth, TX). After staining the posterior hyaloid with triamcinolone acetonide, posterior vitreous detachment was induced, when not already present. The ILM was stained with MembraneBlue Dual (TrypanBlue 0.15% + Brilliant Blue G 0.025%, DORC, Zuidland, The Netherlands) and peeled using a pinch and peel technique. Tamponade was performed with 20% sulfur hexafluoride gas. After the surgery, patients were instructed to maintain a face-down position for at least 3 days. All surgical videos were retrospectively reviewed to assess the

ILM peeling extent and confirm patients' allocation in the two groups.

Retinal Displacement Analysis

Spectral-domain macular optical coherence tomography was performed with the Spectralis (Heidelberg Engineering, Carlsbad, Germany) in all patients. The tangential retinal displacement was evaluated by comparing near-infrared images acquired before and 6 months after the surgery. First, the two images were aligned by lateral translation and rotation using the optic disk as a reference (Figure 1A). Then, vessels were manually segmented by a masked reader (Figure 1B). Finally, the displacement between the two images was computed using the Farneback two-frame motion estimation method.⁴ A vector analysis was performed over the whole image and in the regions of interest defined by a modified Early Treatment Diabetic Retinopathy Study grid dividing the macular region into two concentric 4-mm and 8-mm circles and four quadrants (Figure 1C). A positive horizontal component of the vector indicated a movement toward the disk, a positive vertical component a movement toward the superior retina. Image analysis was performed using a custom Python algorithm.

Statistical Analysis

Statistical analysis was performed using R (version 4.0.0) and RStudio (version 1.2.5042) software. The Wilcoxon signed-rank test was used to compare the magnitude of retinal displacement in patients in the small versus large ILM peeling group. The Mann-Whitney test was used to compare continuous variables before and after the surgery. The correlations among the magnitude of the displacement and other continuous variables were evaluated using Spearman correlation analysis. A *P* value of <0.05 was considered statistically significant.

Results

A total of 44 patients undergoing pars plana vitrectomy for full-thickness MH were included in the study. Demographical and baseline clinical characteristics are reported in Table 1. No significant differences in baseline characteristics were observed between patients in the small and large peeling group (all *P* > 0.05).

Retinal Displacement

Figure 2 shows the retinal displacement analysis in four representative patients. An average overall



Fig. 1. Digital image analysis for the assessment of retinal displacement. **A.** Consecutive near-infrared images are overlapped and colored in red (preoperative image) and green (postoperative one). **B.** The same images after segmentation of the retinal vessels. **C.** Analysis of the optical flow through the Farneback algorithm. White arrows represent the magnitude and direction of retinal displacement; the black lines delimitate the regions of interest consisting of a central 4-mm circle and an outer perimacular 8-mm ring, both divided in four quadrants.

displacement toward the optic disk with a magnitude of $31.3 \pm 22.8 \mu\text{m}$ was observed (X- and Y-components: 13.4 ± 30.4 and $-3.6 \pm 20.0 \mu\text{m}$, respectively). As shown in Table 2, the displacement was 64.9 ± 34.1 in the inner 4-mm circle (X- and Y-components: 25.7 ± 45.1 and $-9.9 \pm 28.2 \mu\text{m}$, respectively) and 50.6 ± 26.7 in the outer 8-mm ring (X- and Y-components: 22.9 ± 32.4 and $-0.1 \pm 26.1 \mu\text{m}$, respectively). The vector analysis of the retinal displacement in the two regions of interest is shown in Figure 3.

Patients undergoing a large peeling had a significantly higher overall displacement ($P = 0.008$), displacement in the inner 4-mm circle ($P < 0.001$) and displacement in outer 8-mm ring ($P = 0.001$) compared with patients undergoing a small peeling (Table 2). The overall displacement did not correlate with the minimal hole diameter ($P = 0.627$), basal diameter ($P = 0.932$), MH stage ($P = 0.738$), or final logMAR best-corrected visual acuity (BCVA) ($P = 0.382$).

Clinical Outcomes

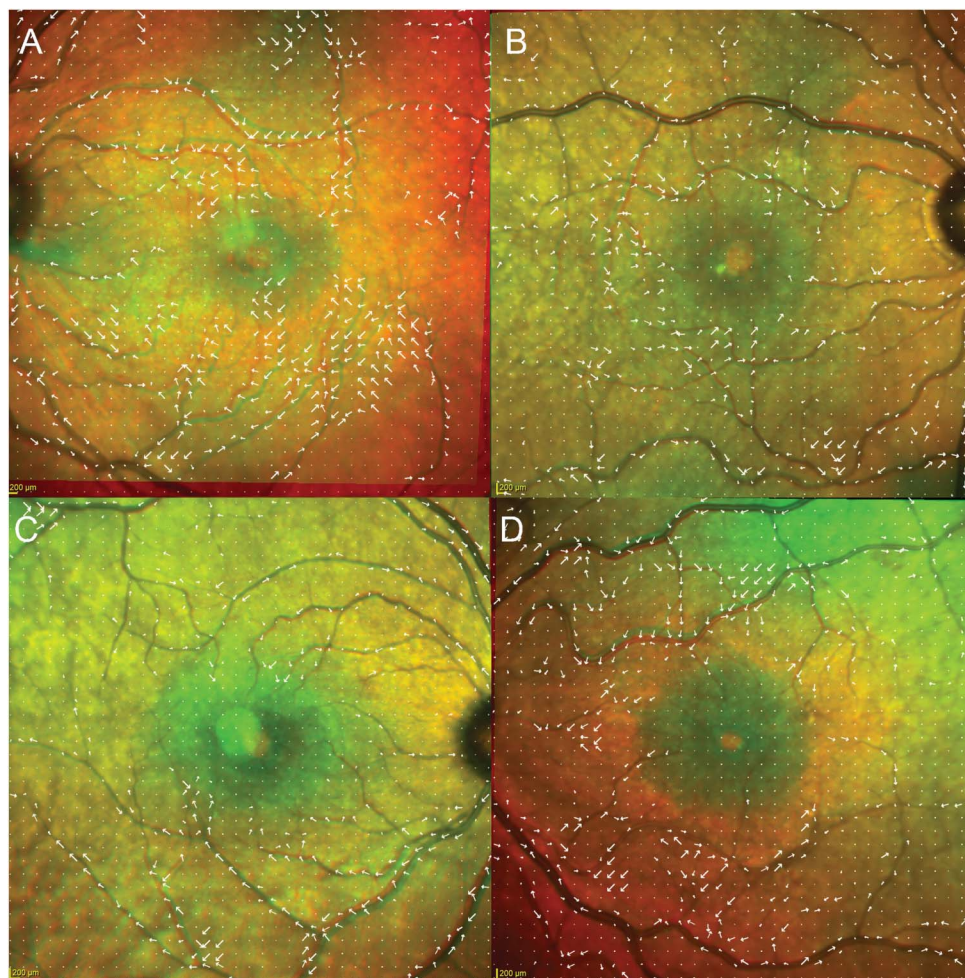
The overall MH closure rate was 90.9%. Macular holes closed in all patients (100%) in the large peeling group, and in 20 of 24 (83.3%) in the small peeling group ($P = 0.055$). Mean BCVA significantly improved from 0.80 ± 0.32 logMAR (20/126 Snellen) to 0.45 ± 0.21 logMAR (20/56 Snellen) after the surgery ($P = 0.001$). The final BCVA was significantly higher in patients in the large versus small peeling group (0.35 ± 0.28 vs. 0.55 ± 0.29 logMAR; 20/45 vs. 20/71 Snellen; $P = 0.042$). However, after excluding the patients who failed, no differences in BCVA were observed (0.35 ± 0.28 vs. 0.48 ± 0.27 logMAR; 20/45 vs. 20/60 Snellen; $P = 0.138$). Postoperative adverse event rates are reported in Table 3. Similar rates of swelling of the arcuate nerve fiber layer and dissociated optic nerve fiber layer were observed in the two groups (both $P > 0.05$). Epiretinal membrane formation was significantly more common in eyes undergoing small ILM peeling ($P = 0.016$). No other postoperative complications occurred.

Table 1. Demographical and Baseline Clinical Characteristics of Patients Treated With a Large Versus Small ILM Peeling

Characteristics	All Patients (n = 44)	Large Peeling (n = 20)	Small Peeling (n = 24)	P*
Women, n (%)	25 (56.8)	10 (50.0)	15 (62.5)	0.404
Age (years)	68.9 ± 8.6	69.2 ± 9.9	68.6 ± 7.6	0.612
Pseudophakic, n (%)	20 (45.5)	8 (40.0)	12 (50.0)	0.507
BCVA (logMAR)	0.80 ± 0.32	0.73 ± 0.23	0.87 ± 0.39	0.540
BCVA (Snellen)	20/126	20/107	20/148	—
MH minimum diameter (μm)	356 ± 154	316 ± 175	389 ± 130	0.172
MH basal diameter (μm)	955 ± 334	$1,020 \pm 359$	902 ± 308	0.229
MH stage				0.613
Stage II	26 (59.1)	13 (65.0)	13 (54.2)	
Stage III	12 (27.3)	4 (20.0)	8 (33.3)	
Stage IV	6 (13.6)	3 (15.0)	3 (12.5)	

*Large versus small peeling (Wilcoxon signed-rank test for continuous variables, Chi-square test for discrete ones).

Fig. 2. Retinal displacement analysis in four representative patients. Consecutive near-infrared images from the same patient are overlapped and colored in red (preoperative image) and green (postoperative one). White arrows represent the magnitude and direction of the average retinal displacement. **A** and **B.** Patients in the large ILM peeling group. **C** and **D.** Patient in the small peeling group.



Discussion

In this study, the tangential retinal displacement occurring in eyes with MHs following pars plana vitrectomy and ILM peeling was evaluated as the

optical flow of the retinal vasculature in consecutive near-infrared images. A retinal displacement toward the optic disk was observed after the surgery, in agreement with previous studies measuring the linear distance between vessel crossings and the optic disk.^{10,11}

Table 2. Mean Retinal Displacement (μm) in Different Regions of Interest in Patients Treated With a Large Versus Small ILM Peeling

Region of Interest	All Patients	Large Peeling	Small Peeling	<i>P</i> *
Whole image	31.3 $\pm$ 22.8	39.6 $\pm$ 26.4	24.4 $\pm$ 17.1	0.008
Inner circle	64.9 $\pm$ 34.1	83.2 $\pm$ 35.0	49.6 $\pm$ 25.0	<0.001
Inferior	69.9 $\pm$ 38.0	88.4 $\pm$ 41.1	54.6 $\pm$ 27.6	0.004
Temporal	56.5 $\pm$ 54.8	78.0 $\pm$ 65.7	38.6 $\pm$ 36.3	0.022
Superior	79.1 $\pm$ 53.2	96.9 $\pm$ 53.9	64.3 $\pm$ 48.8	0.032
Nasal	45.6 $\pm$ 33.8	61.3 $\pm$ 32.1	30.0 $\pm$ 26.9	0.001
Outer ring	50.6 $\pm$ 26.7	64.8 $\pm$ 29.5	38.8 $\pm$ 17.1	0.001
Inferior	68.8 $\pm$ 41.2	87.6 $\pm$ 53.2	53.2 $\pm$ 28.9	0.010
Temporal	43.4 $\pm$ 32.0	51.7 $\pm$ 36.4	36.6 $\pm$ 26.8	0.201
Superior	56.4 $\pm$ 48.1	74.7 $\pm$ 60.5	41.1 $\pm$ 27.6	0.004
Nasal	27.9 $\pm$ 17.0	31.8 $\pm$ 15.0	24.7 $\pm$ 18.3	0.092

*Large versus small peeling (Wilcoxon signed-rank test).

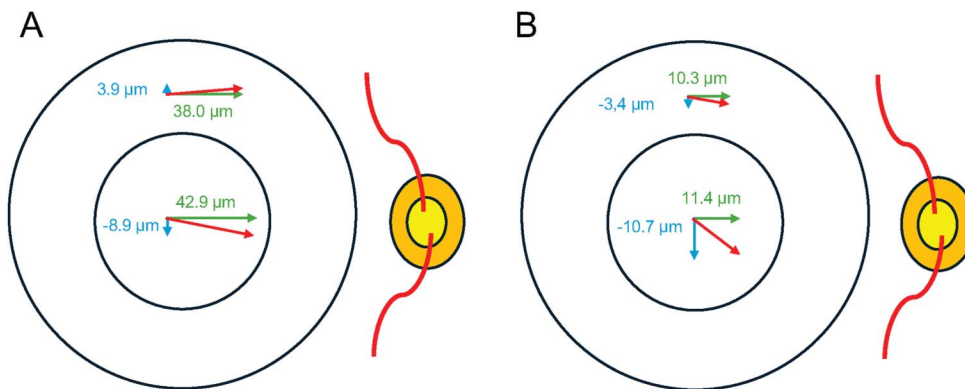


Fig. 3. Vector analysis of the retinal displacement in the central 4-mm circle and outer 8-mm ring. Vectors of displacement are represented in red, the horizontal components in green, and the vertical components in blue. **A.** Large peeling group. **B.** Small peeling group.

The tangential displacement was more pronounced in the temporal retina, and this movement gradient at the margins of the hole might have contributed to its closure.¹¹

The mechanisms underlying the retinal movement toward the disk after ILM peeling are not completely understood. The presence of the rigid ILM might counteract the tractional forces resulting from the intrinsic tension and elasticity of the retinal tissue.¹² Because nerve fibers are anchored at the optic disk, the release of vitreoretinal traction would make the retina retract in that direction. The lower retinal displacement observed in the perimacular area compared with the central macular region might depend on the higher density of large blood vessels, which are among the main determinants of the retinal stiffness.¹³ Compared with the overall displacement of approximately 150 μm reported 6 months after epiretinal membrane surgery,⁸ the retinal movement after ILM peeling was lower. This result was not surprising because the ILM, in contrast to epiretinal membranes, has no contractile properties. As previously noted for epiretinal membrane surgery,^{8,14} the retinal displacement after ILM peeling may not imply the anatomical restoration of the original retinal position but rather a novel equilibrium resulting from a different balance of forces.

There is currently no consensus on the optimal extent of ILM peeling during MH surgery. Although a more complete peeling might theoretically facilitate

the closure of the hole, this advantage must be weighed against the concerns for Müller cells damage, retinal nerve fiber layer thinning, and microscotomas, all of which have been associated with ILM peeling.^{15–17} Aniseikonia represents another potential concern related to a larger peeling area. Indeed, in eyes with rhegmatogenous retinal detachment, the greater tangential displacement occurring after pars plana vitrectomy is associated with worse aniseikonia compared with pneumatic retinopexy.¹⁸ However, a randomized controlled trial from Yao et al¹⁹ reported improved anatomical and visual outcomes with a more extensive ILM peel during MH surgery. Another study from Bae et al²⁰ indicated that a larger ILM peeling was beneficial in improving postoperative metamorphopsia. Nonetheless, other studies reported no anatomical or functional benefits of a larger peeling.^{21,22}

In this study, a greater retinal displacement was observed in eyes undergoing a more extensive ILM peeling. Interestingly, enlarging the peeling up to the vascular arcades did affect not only the perimacular area but also the central macular region. This seems to imply that the tangential forces resulting from the ILM rigidity can be transmitted over distant retinal regions. Thus, relieving the traction in perimacular zone can affect the foveal region, possibly promoting the MH closure. Conversely, as shown in Figure 4, residual ILM can act as a scaffold for cellular proliferation and secondary epiretinal membrane formation. Of

Table 3. Postoperative Adverse Events in Patients Treated With a Large Versus Small ILM Peeling

Characteristics	Large Peeling (n = 20)	Small Peeling (n = 24)	P*
SANFL	2 (10)	2 (8.3)	0.848
DONFL	11 (55.0)	11 (45.8)	0.545
Epiretinal membrane	0 (0%)	6 (25.0)	0.016
Retinal detachment	0 (0%)	0 (0%)	—
Cystoid macular edema	0 (0%)	0 (0%)	—
Endophthalmitis	0 (0%)	0 (0%)	—

*Large versus small peeling (Chi-squared test).

DONFL, dissociated optic nerve fiber layer; SANFL, swelling of the arcuate nerve fiber layer.

note, although a larger peeling significantly reduced the risk of postoperative epiretinal membranes, it did not affect the incidence of swelling of arcuate nerve fiber layer and dissociated optic nerve fiber layer, which are considered optical coherence tomography biomarkers of inner retinal damage.

It should be noted that, in contrast to the image analysis technique proposed by Scarinci et al,⁸ our algorithm was applied to images of segmented retinal vessels rather than unprocessed near-infrared images. Indeed, when comparing near-infrared images, common details such as glares, artifacts, and local differences in luminosity can be erroneously interpreted as retinal movement. Moreover, because a near-infrared-based method cannot be used to infer any information regarding the retinal movement of areas without vessels, those areas were excluded when averaging displacement vectors from different data points.

This article has some limitations that are worth mentioning. First, because this was a retrospective

study, unmeasured variables besides the peeling area could have contributed to the observed differences in the retinal displacement. However, the standardized surgical technique and homogeneity of the population in the two groups likely decreased the chance of unmeasured confounding. Moreover, because an automated method for tracing the retinal vasculature in near-infrared images was unavailable, vessels were manually segmented, and this might constitute a bias. Finally, although a larger ILM peeling was associated with a greater retinal displacement, the difference in the MH closure rate did not reach statistical significance. Further studies with a larger sample size are still required to assess whether that the extent of retinal displacement may have a clinically meaningful impact on surgical success rates.

In conclusion, the optical flow of the retinal vasculature may be employed to measure the tangential retinal displacement occurring following MH surgery. A larger extent of the ILM peeling is associated with a greater retinal movement, possibly leading to improved surgical success rates.

Key words: retinal tectonics, retinal displacement, macular hole, vitreoretinal surgery.

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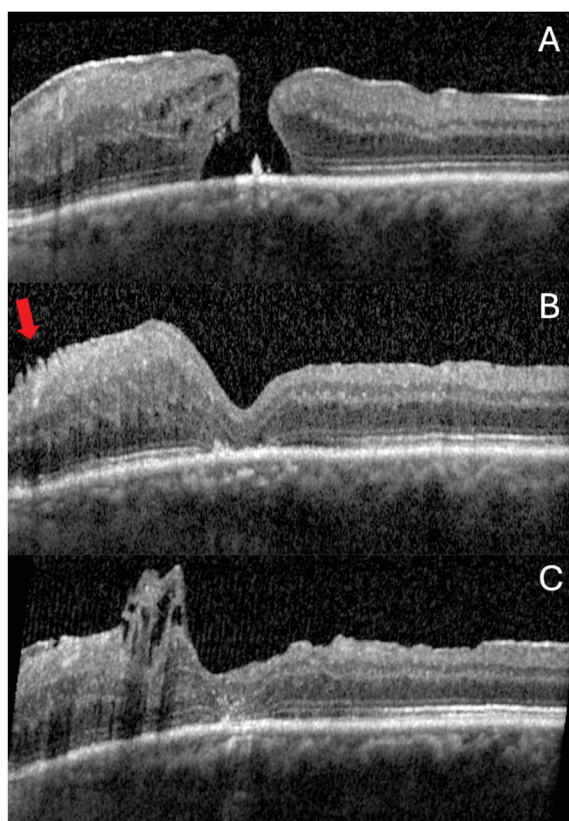


Fig. 4. Macular optical coherence tomography of a representative patient who underwent vitrectomy with small ILM peeling. **A.** The preoperative scan showed a full-thickness MH. **B.** One month after the surgery, the MH closure was closed, and wrinkles of the unpeeled residual ILM appeared on the temporal side of the optic disk (red arrow). **C.** Six months after the surgery, contraction of the residual ILM progressed with the formation of an epiretinal membrane. The patient developed significant metamorphopsia.

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